Supplemental Information for Data-driven recommendations for increasing reliability in measures of infant word recognition

Contents

1 Data description

Figure S1 shows the distribution of ages in the Peekbank data used in this paper.

Figure S2 shows the distribution of AOIs for different datasets and different timepoints. Some datasets have less data at the start or end due to varying length recordings, or being a mix of multiple experiments with different recording windows. Most (but not all) datasets had data before the point of disambiguation and up to 3000ms. Datasets varied dramatically in overall size.

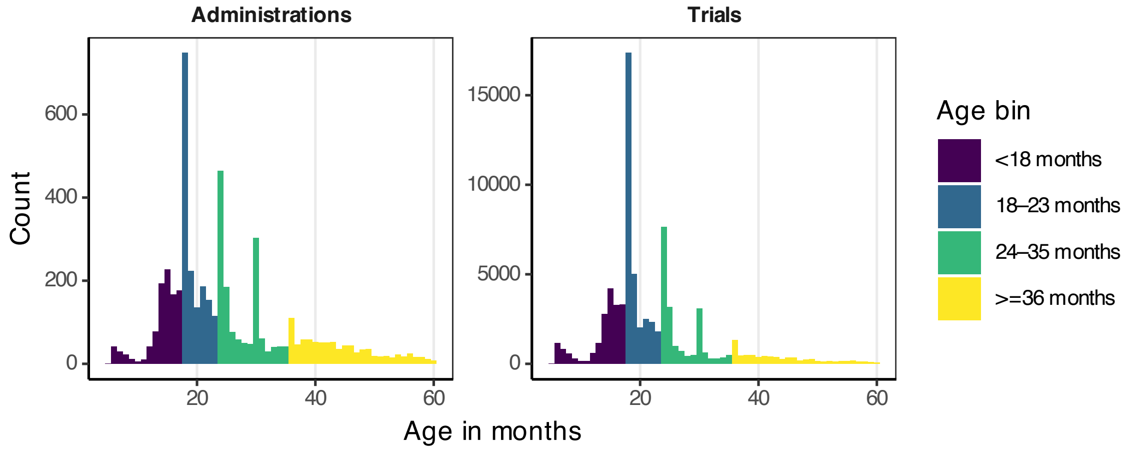

Figure S1: Age distribution for trials and administrations, with the age bins used for analysis.

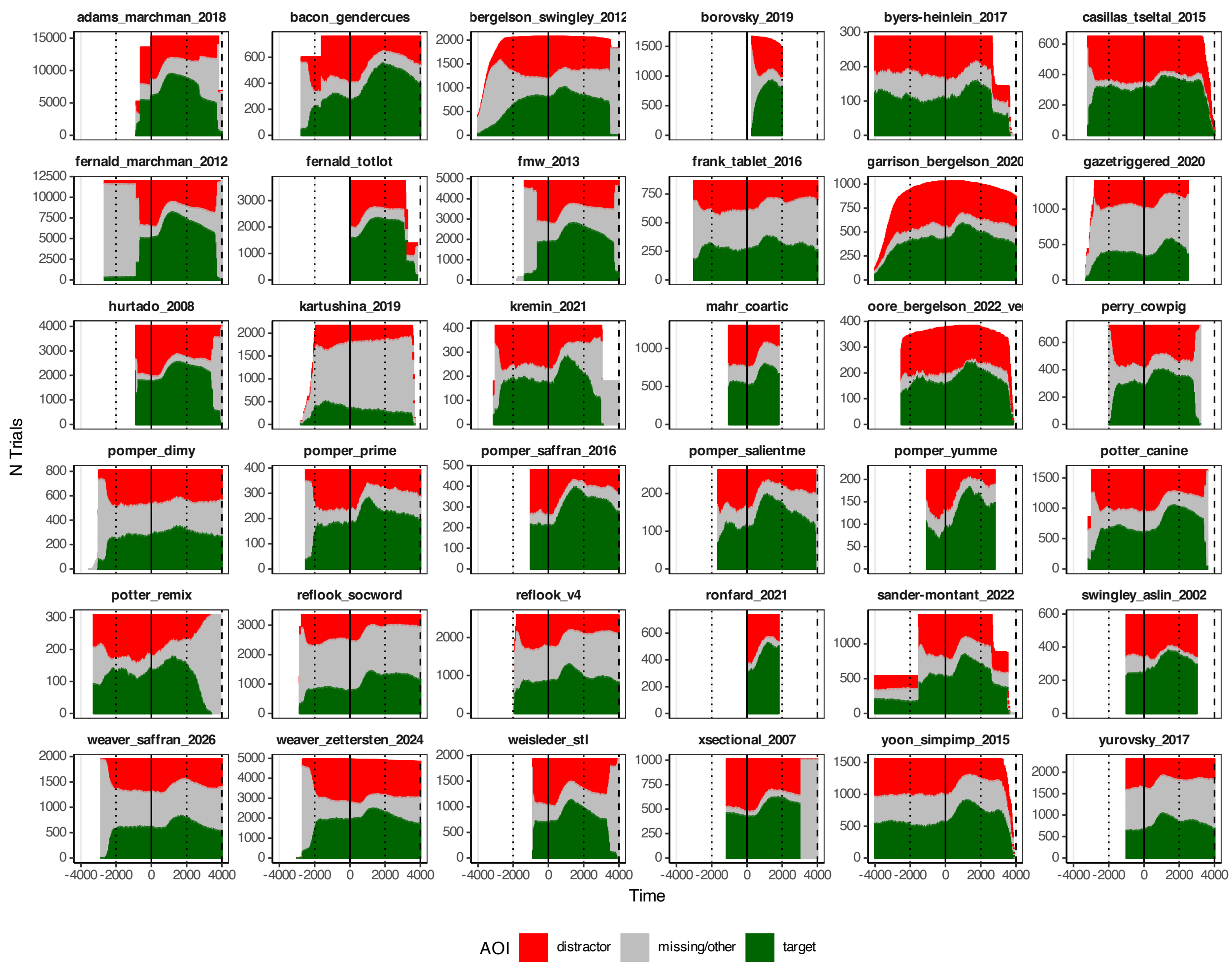

Figure S2: Distribution of AOIs across timepoints and datasets. Solid vertical line at point of disambiguation, Dashed lines at 4000ms, and dotted line at -2000ms and 2000ms.

2 Methods details

2.1 Criteria for inclusion

The "vanilla" status of a trial is coded in the Peekbank dataset. As part of the 2026.1 release, we implemented an updated set of "vanilla" criteria. Multiple coders systematically inspected the coding of each dataset to identify vanilla trials and ensured that the application of the vanilla criteria was consistent across all datasets. Cases where there was potential ambiguity were discussed and documented in the README associated with each dataset, to ensure that the criteria used were consistent and explainable. The criteria used were as follows:

Auditory Stimulus:

- familiar words/ target word is a real noun (no part words)
- target word is the first point of disambiguation
- no duplicated target words (i.e., repeating the target word multiple times was deemed non-vanilla)
- well-formed and grammatical carrier phrase (there were no restrictions on carrier vocabulary, and the target NP could appear anywhere in the sentence.)
- no informative structure prenominally or postnominally (informative verb or adjective; an uninformative verb or adjective is OK)
- no nonsense words anywhere (including the carrier phrase)
- no language/ speaker/ accent switching within the trial
- no intentional background noise or audio filtering
- no additional restrictions on noun difficulty

Visual stimuli:

- both target and distractor images are familiar objects (i.e., each could be valid familiar targets; non-prototypical objects are OK, but unnatural/ artificially modified exemplars are not)
- no novel objects are visible
- The target referent is the focal object of its image (dog on grass for dog is OK; apple on plate is OK; apple and orange for orange is not, because both the apple and orange are competing for focal attention).

2.2 Details about ICC

A number of different correlation metrics are all abbreviated as ICC. In this study, we use the ICC for *inter*-rater reliability for a "random factorial model without interaction" measuring consistency and looking for the reliability of an average of raters. We use the implementation in the "agreement" package (Girard 2019), but we are not aware of this specific combination being fully documented anywhere.

This ICC looks at the ratio between by-subject variance (by-subject variance + by-rater variance + error variance)

Here's what this ICC means in comparison to other possible ICCs:

- It is an inter-rater reliability measure, meaning that it measures how well raters (or in our case items) correlate with each other as opposed to with themselves.

- It uses the "random factorial model" (sometimes called Type 2). This is for when all raters rate all subjects, but some raters may rate some subjects more than once. Both raters and subjects are assumed to be sampled from a larger population of interest.

- It is “without interaction”. This variant can be used when there are single ratings for a subject-rater combination (preventing the accurate determination of subject-rater interaction effects), or when subject-rater interactions are not of interest (and can be lumped into the overall error). This version is documented in Gwet (2014) (p. 256). This measures the ratio of by-subject variance / (by-subject variance + by-rater variance + error variance).
- It is a consistency measure rather than an agreement measure. Rather than measure the exact agreement between raters, it instead measures whether the slopes across raters for different subjects are the same. It does not penalize situations where one rater is consistently lower (i.e. a later-learned word). This omits the by-rater variance from the denominator. We are unaware of the combination of random-factorial model and consistency being explicitly documented, but this formula aligns with the discussion of consistency in Gwet (2014) (p. 274).
- It measures consistency on average across a pool of raters by dividing the error term by the number of raters.

The overall formula for ICC is then by-subject variance / (by-subject variance + (error variance / number of raters))

We use the package “agreement” because it, unlike other ICC packages, handles cases where there is missing or uneven data (by using Gwet’s formulations). This is essential because peekbank has uneven data, especially for cases involving RT or post-exclusion data.

We only use the ICC measure to compare across the same datasets with different processing methods applied. We do not interpet the ICC values against any external cutoffs of reliability.

3 Accuracy

Figure S3 shows a version of Figure 1 split by age bin categories. Older children have overall higher ICC reliability, but different ages do not show different regions of high reliability. Figure S4 shows the model estimates and 95% CIs for the effect of different windowing options (comparing to 600–4000 ms as the index condition). Negative numbers indicate that the value is lower in the given window than for the index condition; higher numbers indicate that the value is higher in the given window than in the index condition. Figure S5 shows the model coefficients and 95% CIs for linear mixed effect models run separately for each age bin. While the degrees of difference vary across ages, the overall patterns are the same.

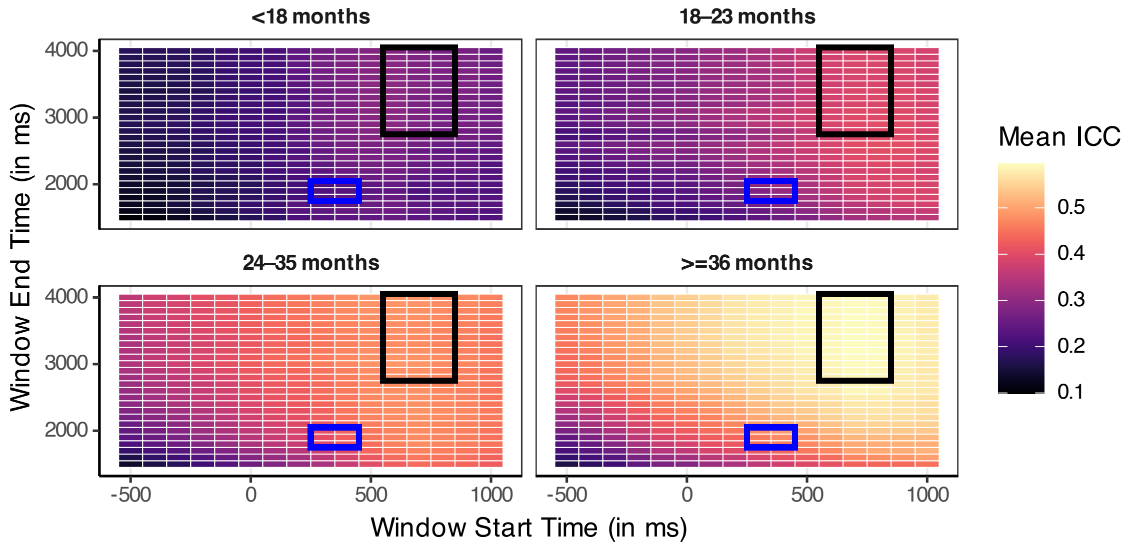

Figure S3: A heat map showing the average ICC across datasets for different start and end points for the accuracy window, split by age bins. A region with the cross-age highest ICC (best reliability) is outlined.

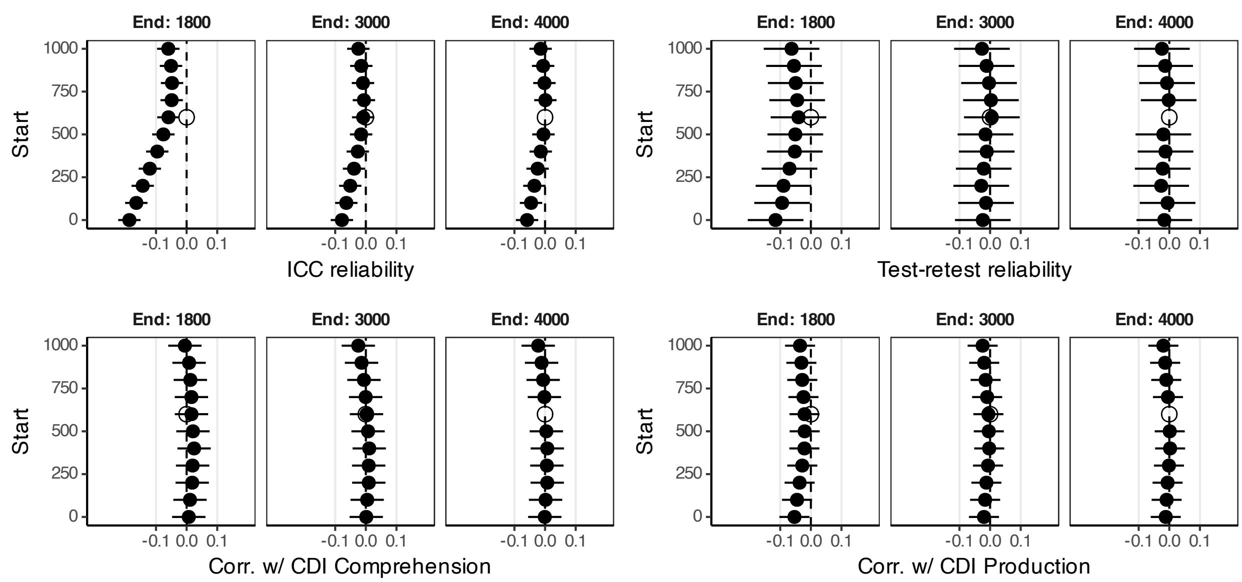

Figure S4: Model results comparing the effects different accuracy windows to the index length of 600–4000 ms for different dependent variables.

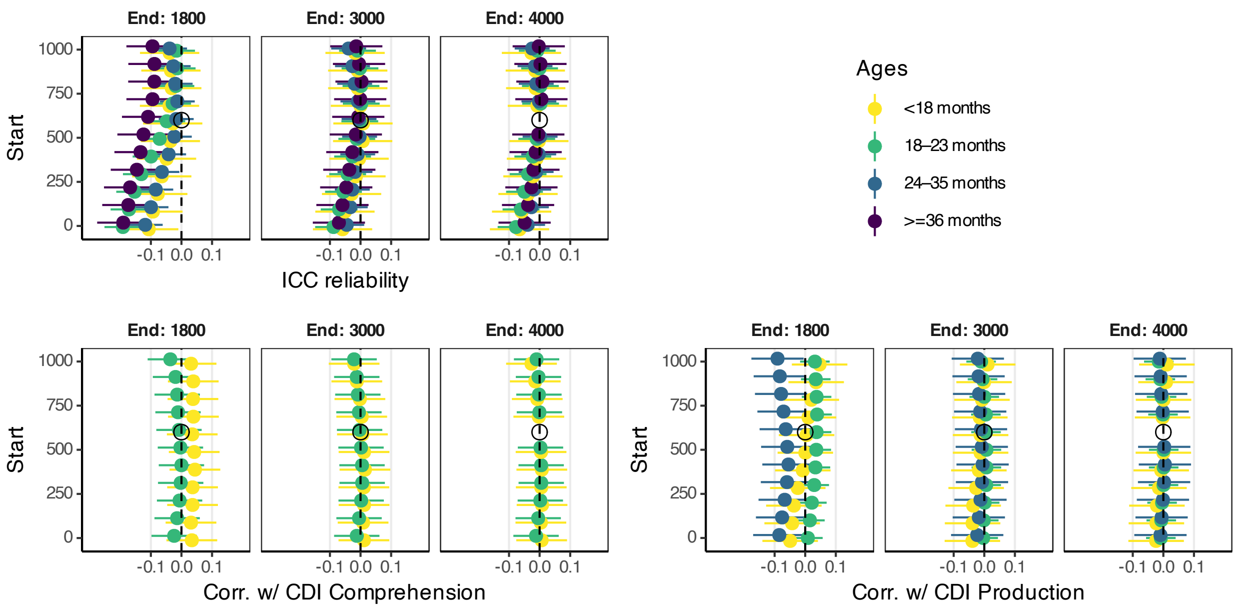

Figure S5: Model results split out by age comparing the effects different accuracy windows to the index length of 600–4000 ms for different dependent variables.

4 Baseline correction

For baseline correction, we only considered datasets where the earliest time point was at or before -2000 ms, to ensure there was adequate per-onset data.

Figure S6 shows how baseline corrected measures compare to normal accuracy measures for a 300–1800 ms window, as an example of generalization. Figure S7 shows model results comparing different baseline correction options to a plain accuracy measure. Baseline correction is substantially worse. Figure S8 shows model results comparing baseline corrected accuracies to the index condition of normal accuracy for each of the age bins. Across age bins, baseline correction is often worse and in no case better than normal accuracy.

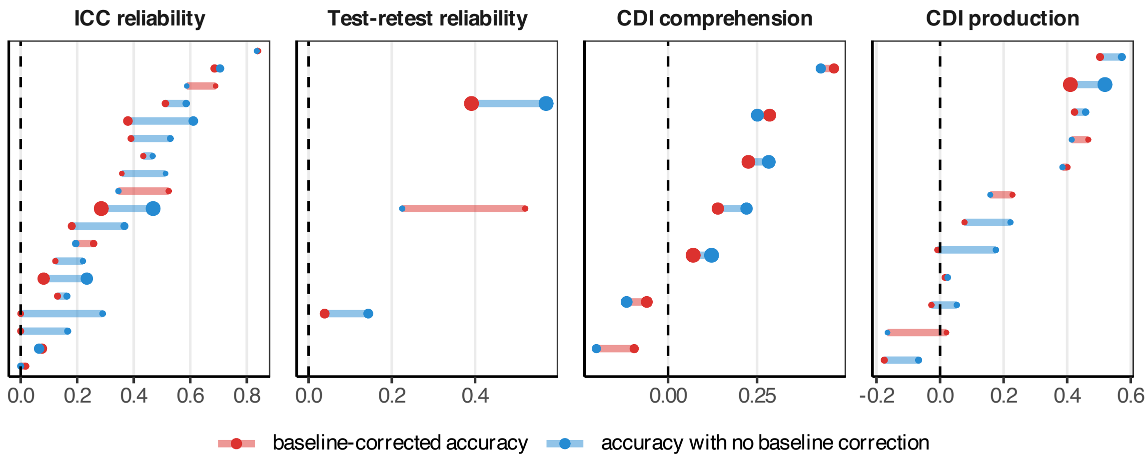

Figure S6: Comparison between a baseline-corrected accuracy in red and accuracy in blue across datasets and measures. Both use a post-stimulus window of 300 - 1800ms, the baseline window was -2000 - 0ms.

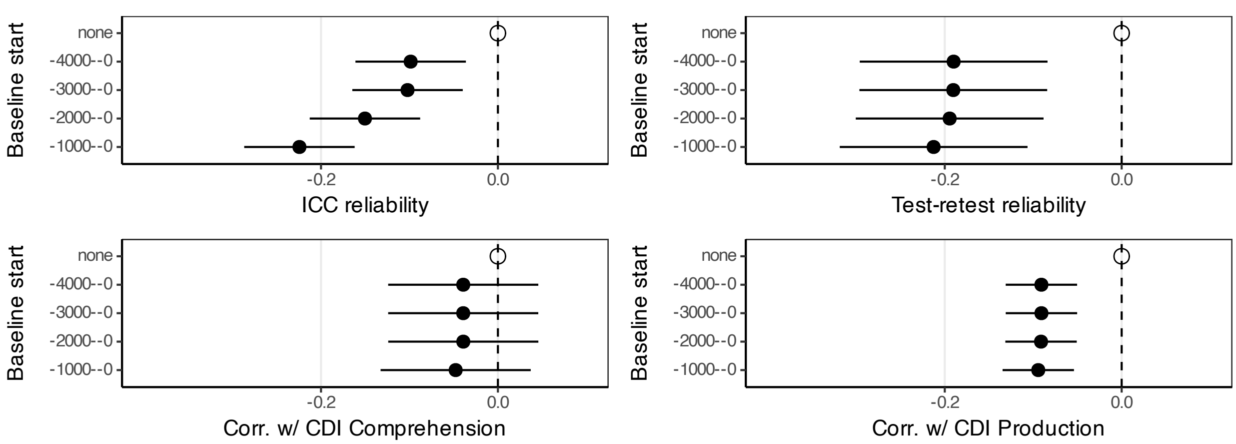

Figure S7: Model results comparing the effect of different baseline windows for baseline correction to uncorrected accuracy. All use a post stimulus window of 600–4000 ms.

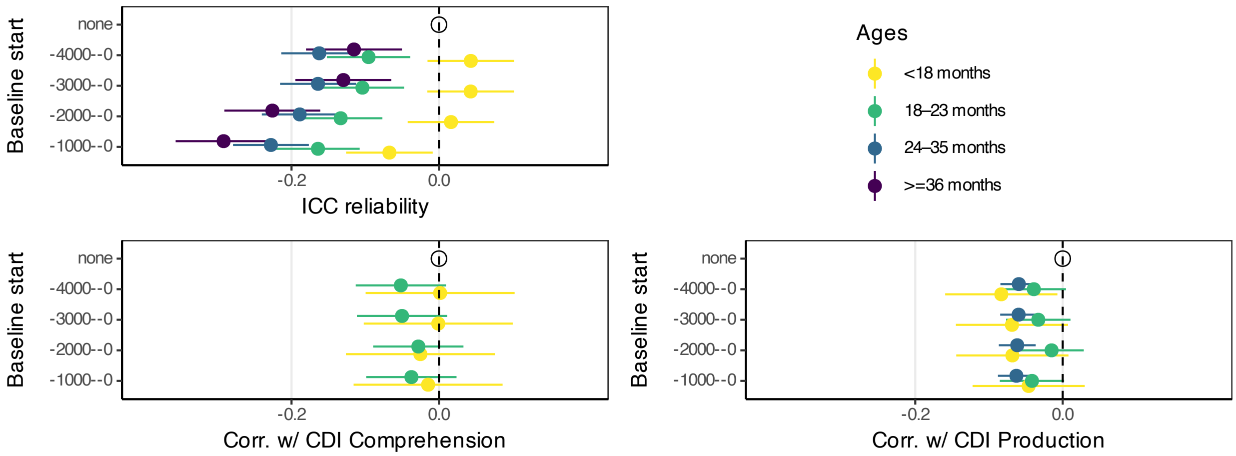

Figure S8: Model results by age group comparing the effect of different baseline windows for baseline correction to uncorrected accuracy. All use a post stimulus window of 600–4000 ms

5 Reaction time

Figure S9 shows a heat map of ICC relability across different RT conditions, split by age. Younger children are less reliable overall, but there are not strong age-specific trends in what measurements are most reliable.

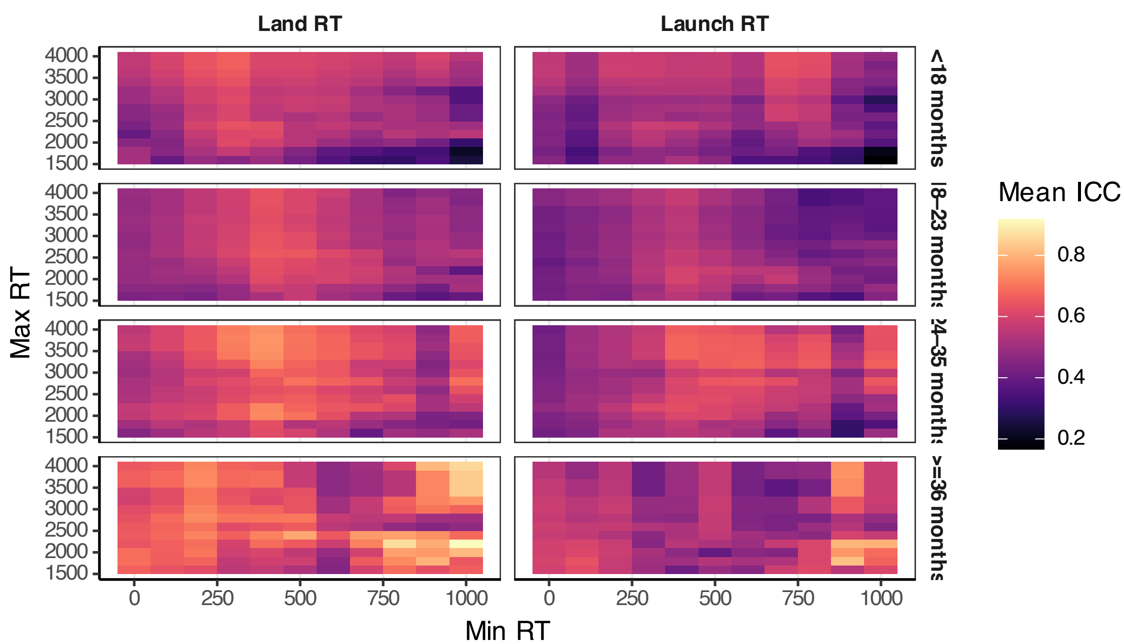

Figure S9: Heat map of different parameter options for measuring RT and how they on average affected ICC.

Figure S10 shows models estimates for different model parameters compared to the index condition of untrimmed log land with 400 minimum RT. Because faster RT tends to correlate with higher language skill and thus higher CDI measures, for the bottom two panels, positive numbers indicate worse correlations in the tested condition compared to index. Figure S11 shows the reliability and validity of different reaction time parameters, split by age bin.

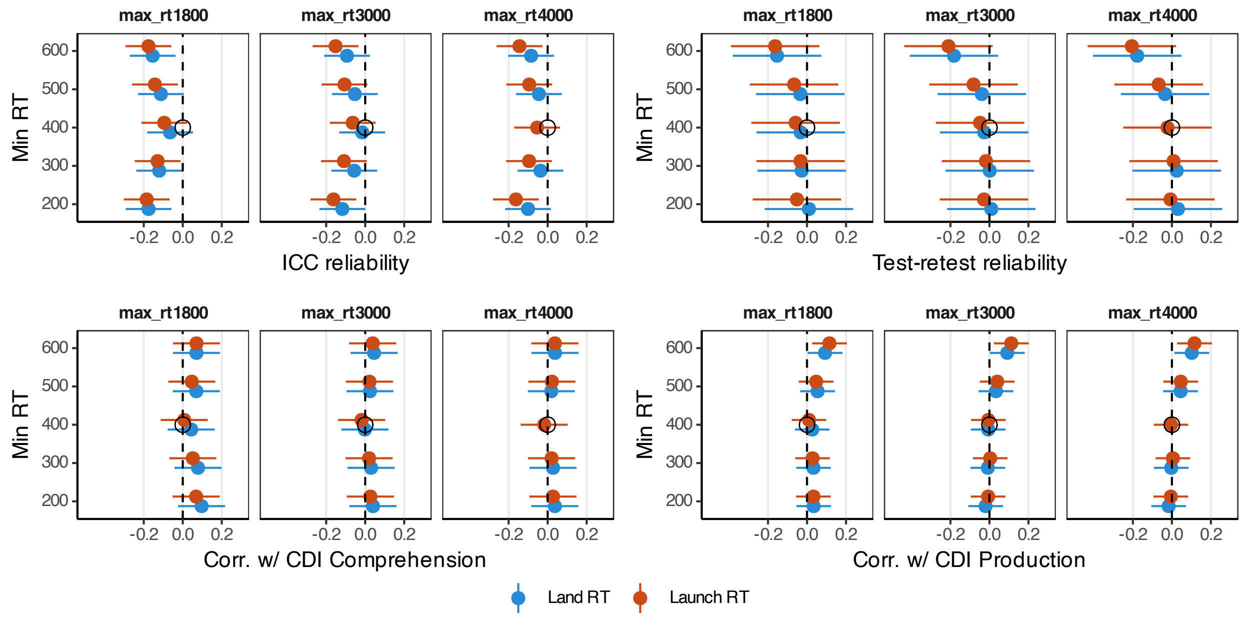

Figure S10: Model results for RT. The index condition is a log Land RT from 400–4000 ms. All RTs are shown log in this figure.

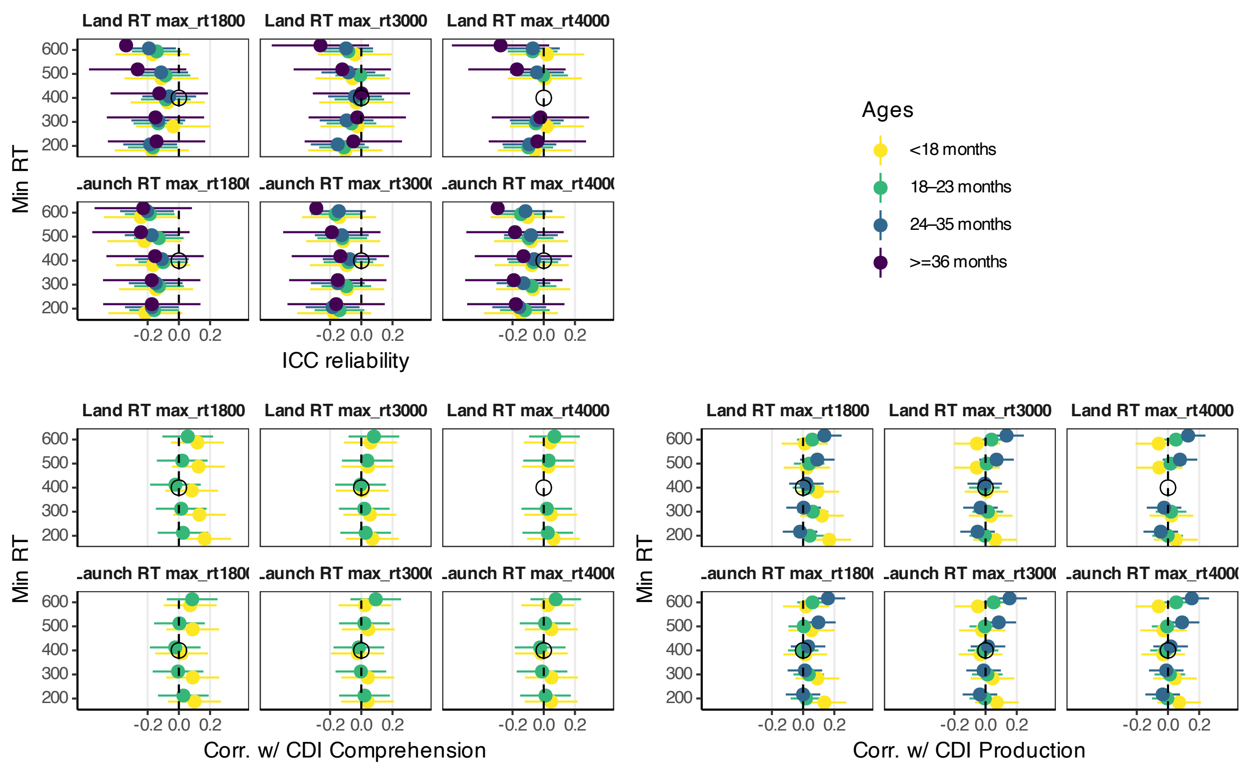

Figure S11: Model results for RT. The index condition is a log Land RT from 400–4000 ms. All RTs are shown log in this figure.

For thoroughness, we also considered other dimensions of RT parameters focused on the 400–4000 ms RT range.

In Figure S12, we consider land versus launch RT, log versus raw, and whether to exclude long shifts. While not formally mentioned, launch based RT is often restricted to RTs where the child's eyes make it to the target in a reasonable amount of time (ex. 600 ms). This restriction on shift-length can also be applied to land based RT. As shown in the figure, these differences in calculation do not make significant differences.

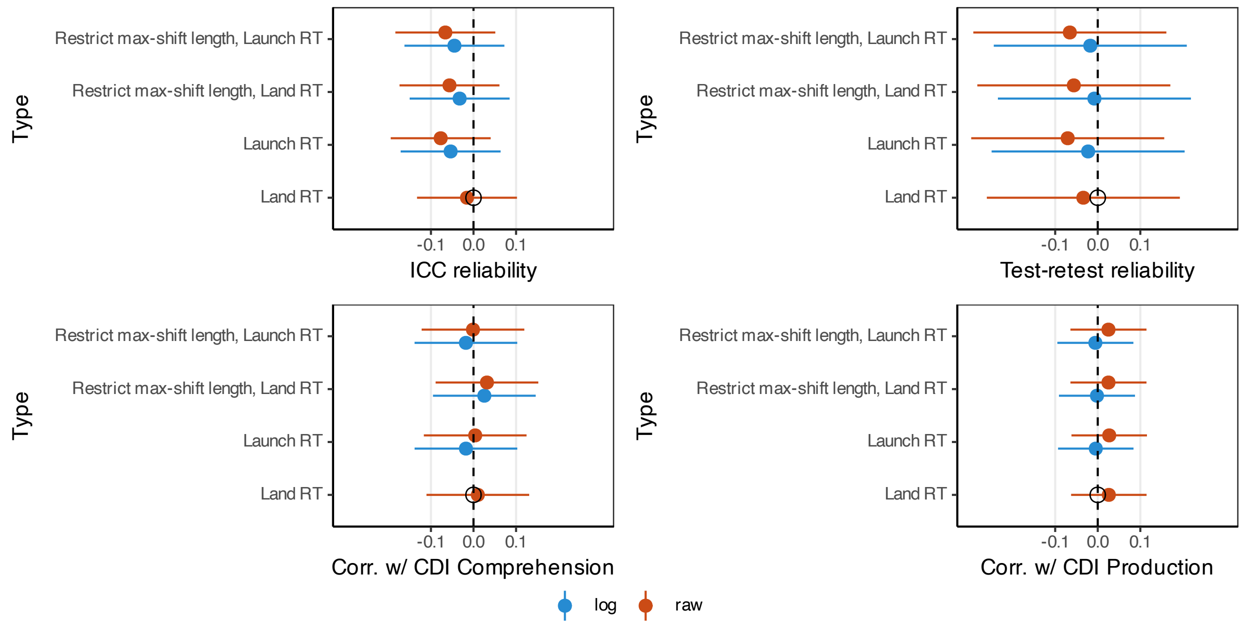

Figure S12: Model results for different ways of computing RT on the 400–4000 ms window, focused on max-shift length and transformation.

6 Exclusions

6.1 Accuracy trial exclusions

Figure S13 shows effects of different trial-level exclusions on outcomes. Figure S14 shows age splits of the same.

Excessive requirements lead to an increase in ICC, but a decrease in other metrics. A requirements to look at the time of auditory stimulus onset just leads to data loss without reliability or validity gains.

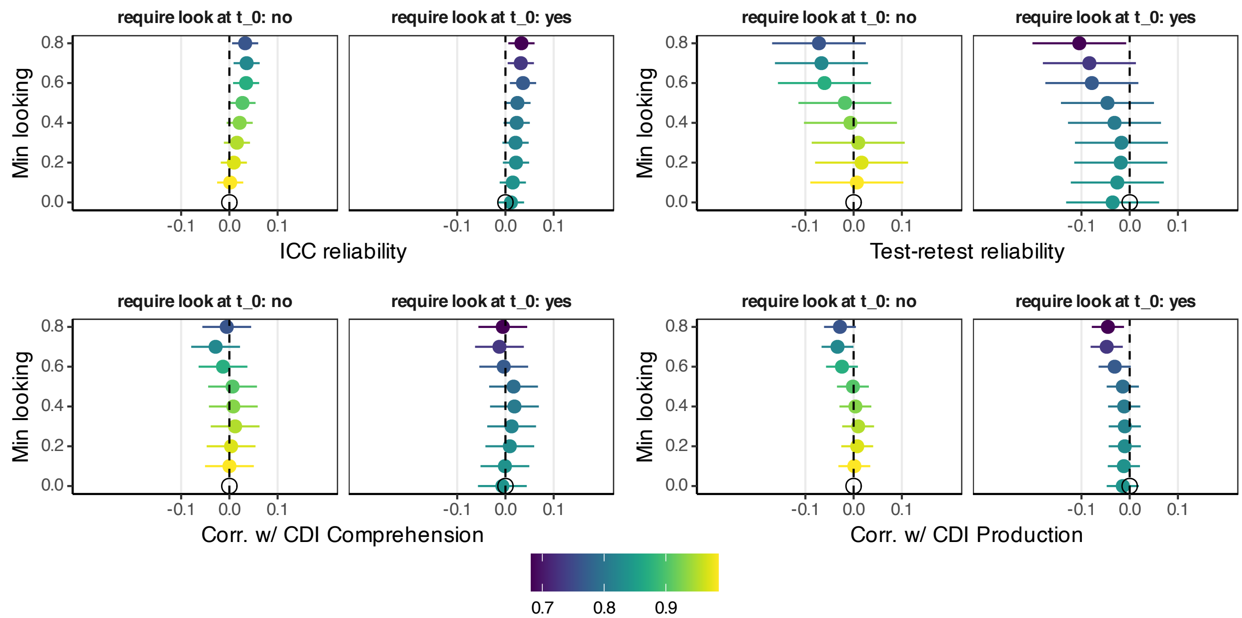

Figure S13: Model results for possible trial-level exclusions, compared to no exclusions. Colors are overall rate of trial-level data retention.

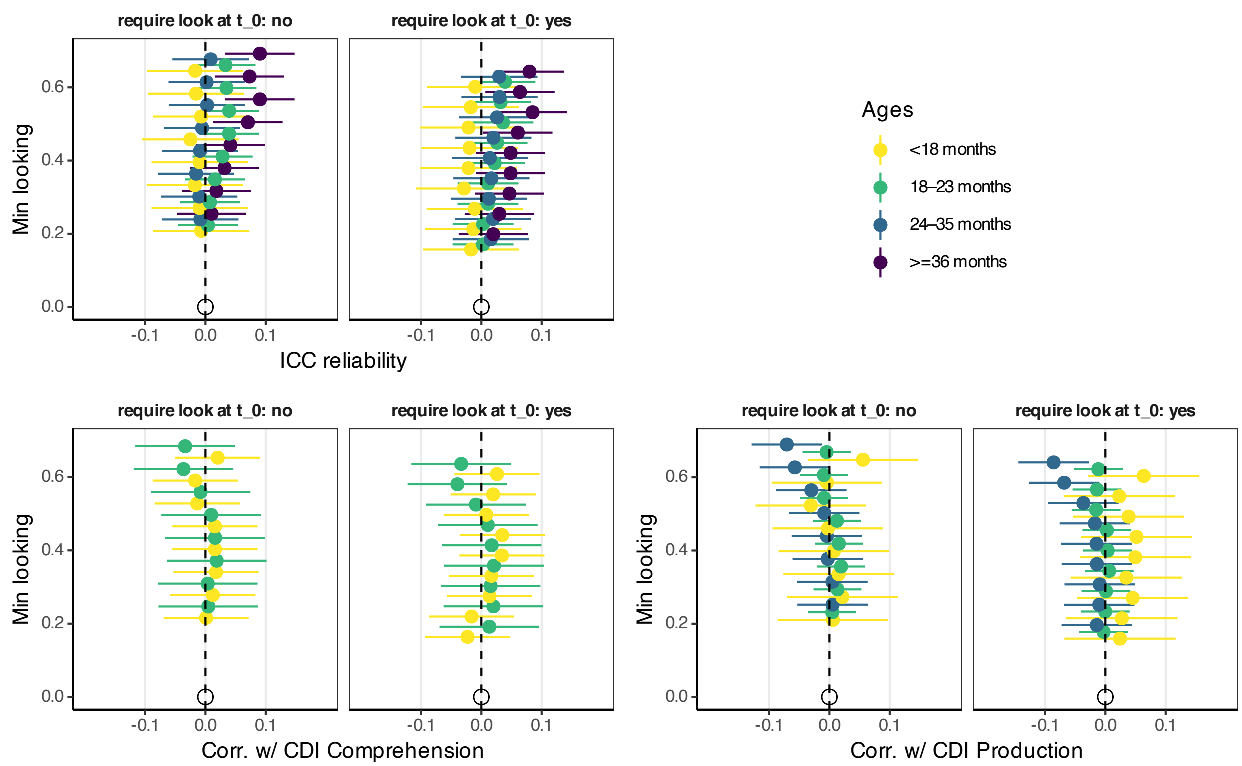

Figure S14: Model results for possible trial-level exclusions for each age group, compared to no exclusions.

6.2 Accuracy exclusions by administration

Target number of total trials is not a peekbank field, so we guessed it by the largest number of trials any child in that dataset had, and adjusted for datasets where it was clear from the distributions there were different numbers of vanilla trials at different ages or administrations.

Overall model results are shown for trial count exclusions in Figure S15 and for trial fraction exclusions in Figure S16. Age splits are shown in Figure S17 and Figure S18.

More trials numerically improves test-retest reliability (at least when there are enough trials), but there are trade-offs against number of retained administrations.

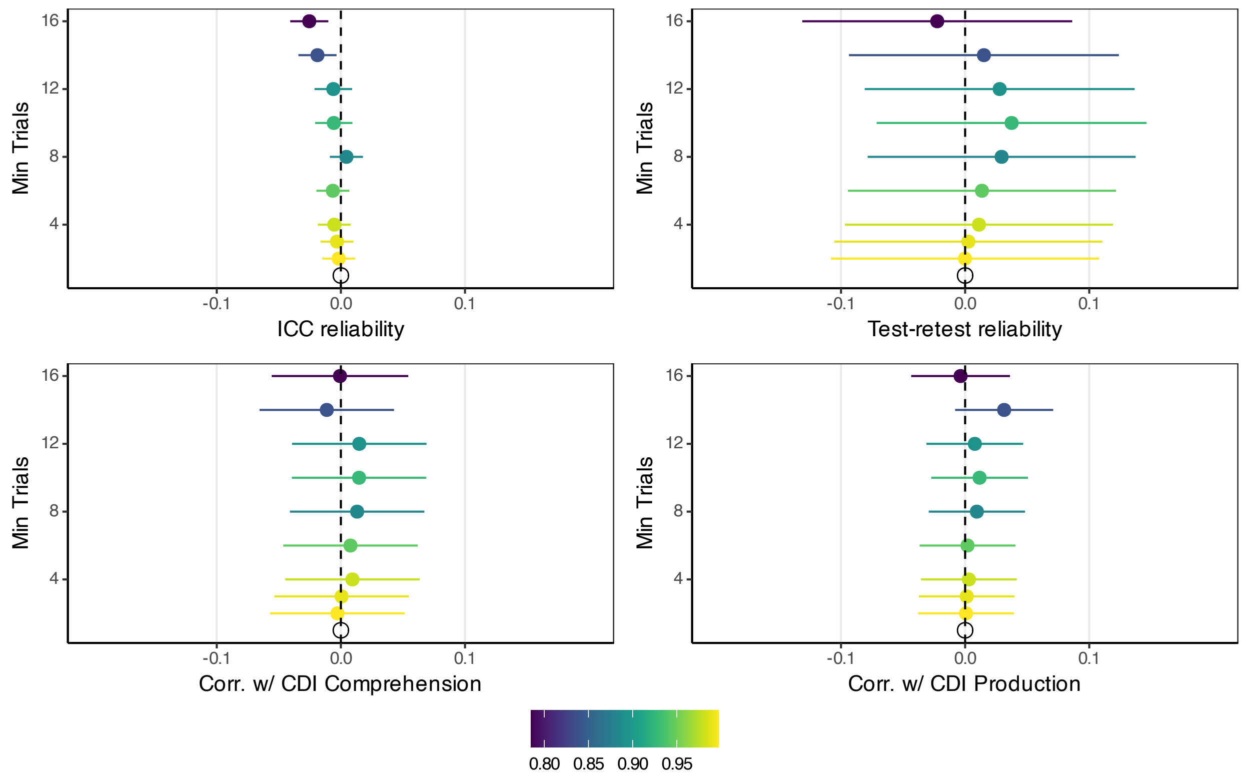

Figure S15: Model results for possible trial count based exclusions, compared to no minimum trial count. Colors indicate fraction of administrations retained.

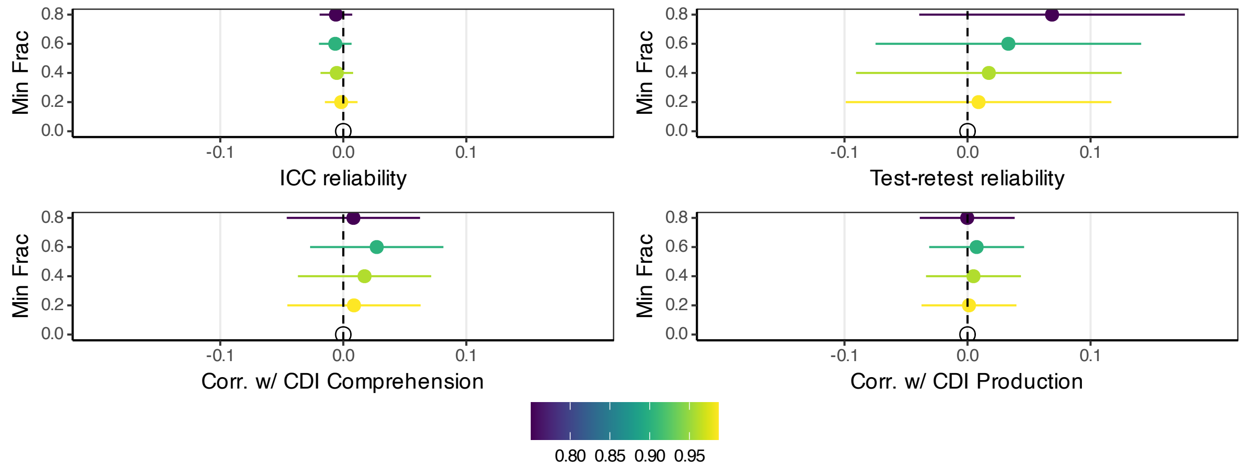

Figure S16: Model results for possible fraction of trial based exclusions, compared to no minimum. Colors indicate fraction of administrations retained.

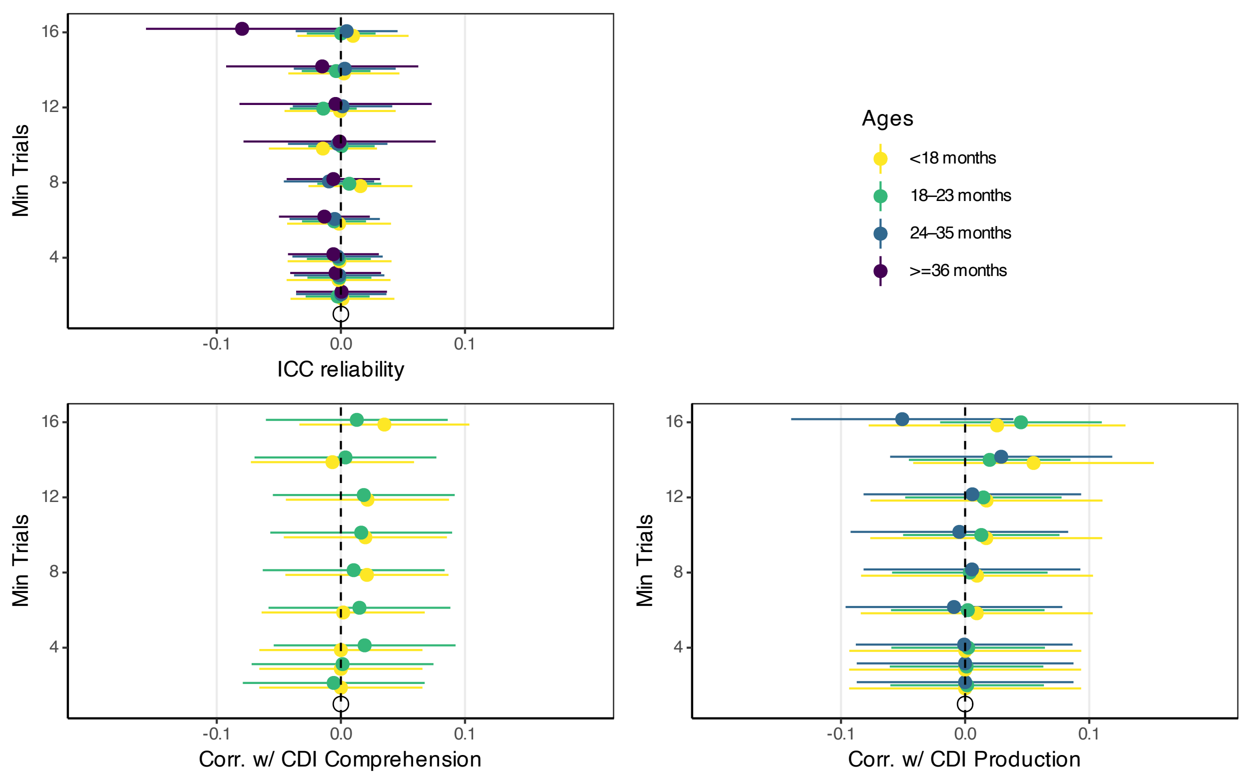

Figure S17: Model results for different age groups for possible trial count based exclusions, compared to no minimum trial count.

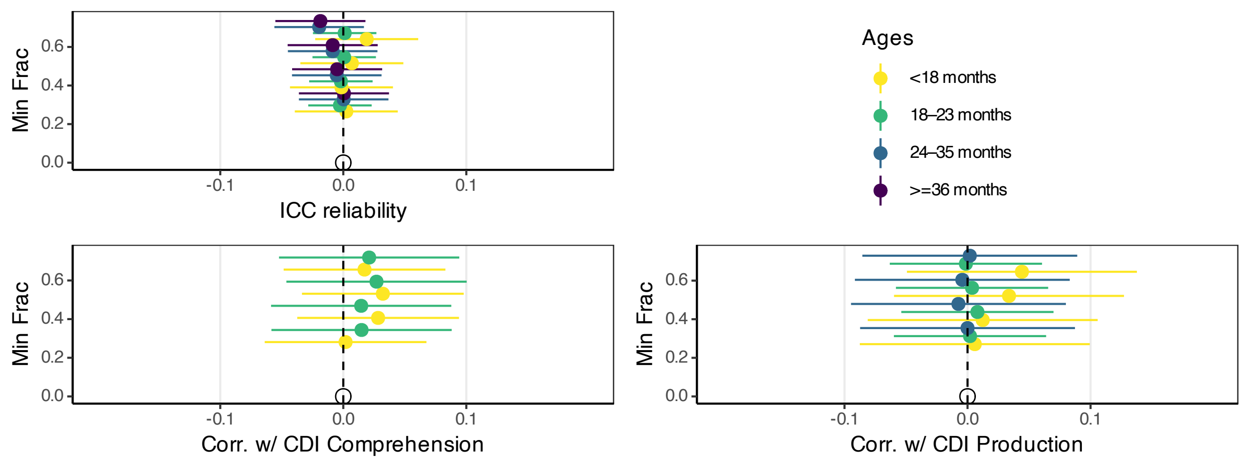

Figure S18: Model results for different age groups for possible fraction of trial based exclusions, compared to no minimum.

6.3 Reaction time exclusions

Excluding children who have fewer valid RT trials may increase test-retest reliability, and has unclear relationships with validity. Overall model results are shown in Figure S19 and model results by age group are down in Figure S20. Minimum trial requirements do help for test-retest reliability, and CDI comprehension, at the cost of reducing the number of included administrations.

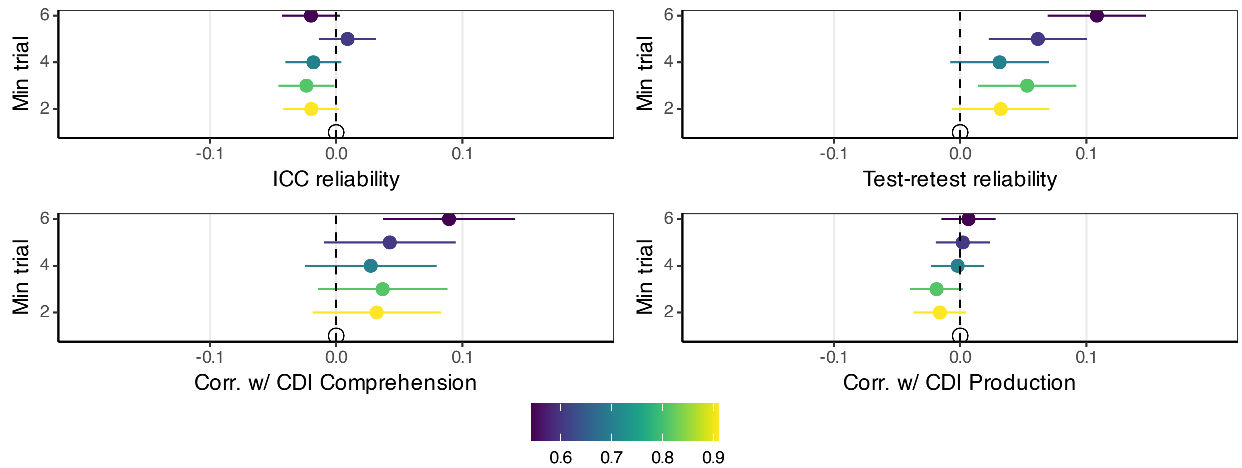

Figure S19: Effects of imposing minimum RT trial requirements, compared to no trial minimum, colored by what fraction of administrations overall are retained.

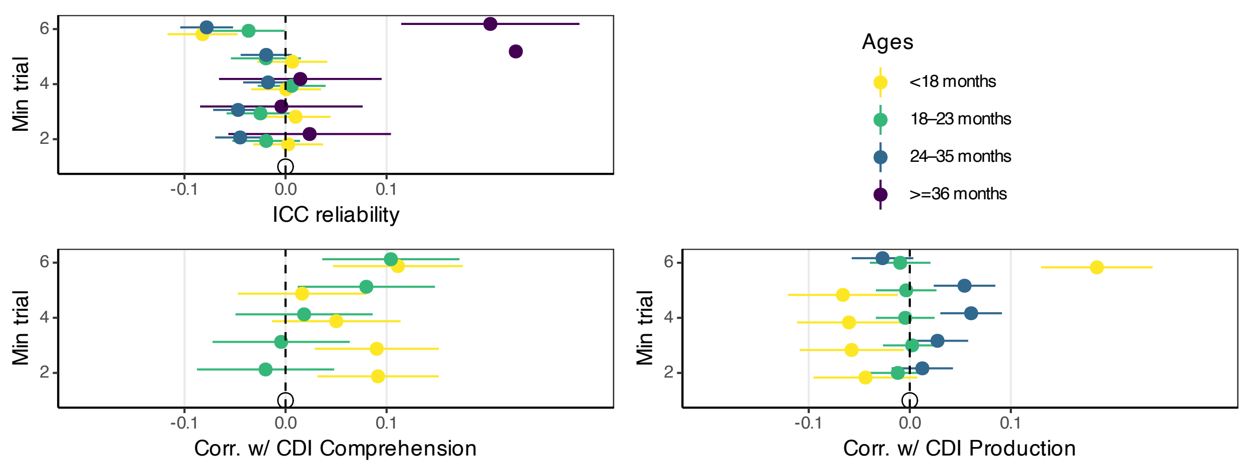

Figure S20: Effects of imposing minimum RT trial requirements, compared to no trial minimum, split by age bin.

7 Order effects

One potential concern, especially when children see a number of trials, is that the repetition of target and distractor images (or just the target nouns) may lead to order effects or preferences for familiarity, i.e., the image that was named last. It is unclear how these effects might manifest, so we looked for possible order effects in a combinatorial fashion.

We considered 5 dependent variables:

- “early looking” on the -2000 to 0 pre-window
- “late looking” on the 0 to 600 window (between onset of stimulus name and when we start an accuracy window)
- accuracy on 600–4000 ms
- RT on 400–4000 ms
- whether there is a valid RT (if children’s looks at 0 are disproportionate, this could reduce the trials where RT is possible)

We also considered a set of independent variables across different models:

- trial order: for stimuli seen for the first time, how does position in order change viewing behavior?
- seen previously (as distractor) – for stimuli, how does being seen for the first time as the target compare to being the target when seen once previously as the distractor? (We restrict to pairs of stimuli that are always each other’s distractors.)
- seen previously (as distractor) x trial order
- seen previously (as target) – for stimuli, how does being seen for the first time as the target compare to being the target when seen once previously as the target? (We restrict to pairs of stimuli that are always each other’s distractors)
- seen previously (as target) x trial order
- seen previously (as either) – for datasets where the same items could be seen previously as either the target or distractor, we do a 3-way comparison between first time viewing and one previous viewing as either target or distractor.
- seen previously (as either) x trial order

We fully crossed the 5 dependent variables with the 7 model predictor structures.

The coefficients from all these models are shown in Figure S21. Effects related to trial order were multiplied by 10 for the effect of a difference in 10 trials for viewability. The effects of having a target image that has been seen 1 time previously (as distractor or target), versus a target image that is being seen for the first time are (if they exist) have central estimates around 1% of the time more or less looking (both for windows around 0 and actual accuracy). These effects may well interact with trial order, although again, effects and interactions, if they exist, are small. Log RT has a larger scale than the other measures that are all on 0–1. Switches to target may be a bit faster if the target was previously seen, but only when interaction with trial order is considered in the model. If anything, there is more looking to previous distractors on the accuracy window. At least across datasets, we cannot detect reliable effects of previously seen items or trial order on any of the dependent variables.

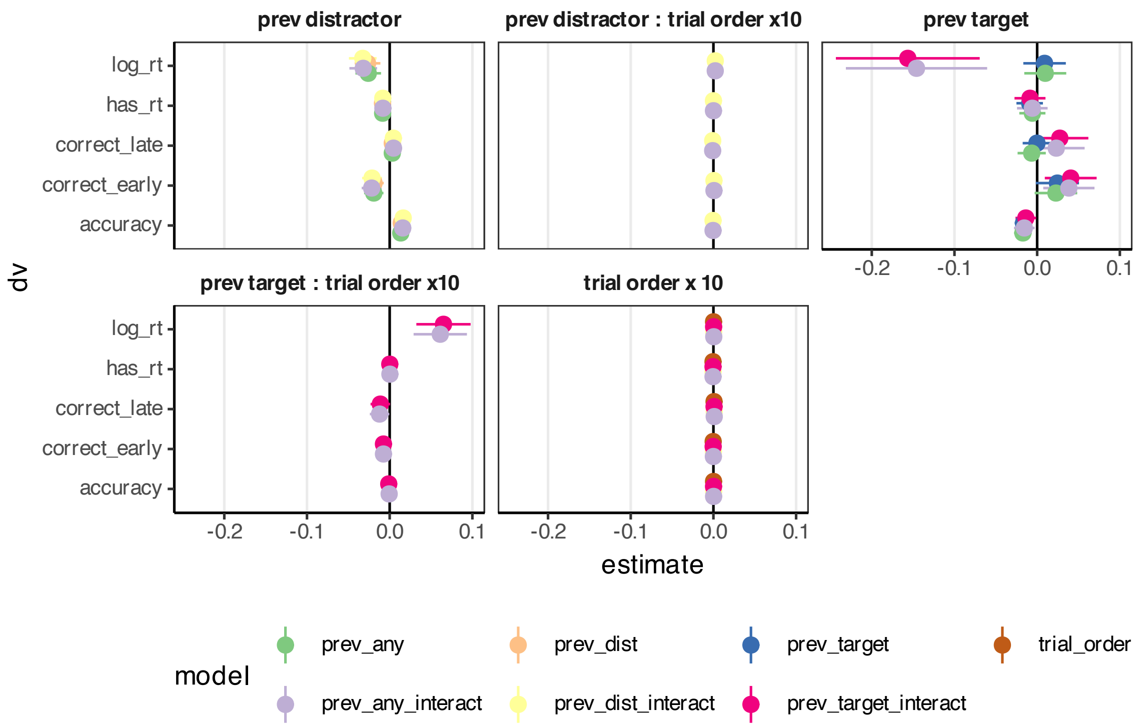

Figure S21: Fitted coefficient estimates from a series of models investigating potential order and repetition effects. For scaling, trial order effects are shown for every 10 trials.

References

Girard, Jeffrey. 2019. *Agreement: R Package for the Tidy Calculation of Inter-Rater Reliability*. https://github.com/jmgirard/agreement.

Gwet, Kilem Li. 2014. *Handbook of Inter-Rater Reliability: The Definitive Guide to Measuring the Extent of Agreement Among Raters*. 4th ed. Advanced Analytics, LLC.